

Radix 2

plain binary — one digit per bit of B

Step 1 — the number

B is an n -bit two's complement value with bits $B^j \in \{0, 1\}$. The top bit counts negative:

$$B = -2^{n-1} B^{n-1} + \sum_{j=0}^{n-2} 2^j B^j$$

Step 2 — the product

$$AB = -2^{n-1} B^{n-1} A + \sum_{j=0}^{n-2} 2^j B^j A$$

$B^j A$ is a selection, not a product:

B^j	$B^j A$
0	0
1	A

n digits, n partial products, one multiple (A). The sign never enters the digits: it is the weight of the top plane, so exactly one row is subtracted.